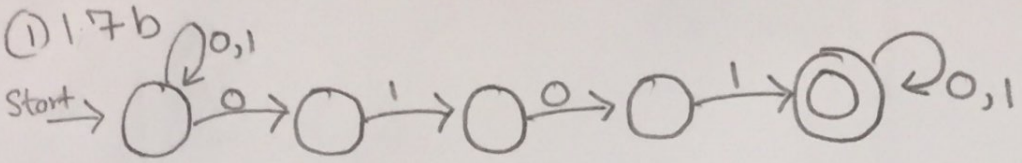
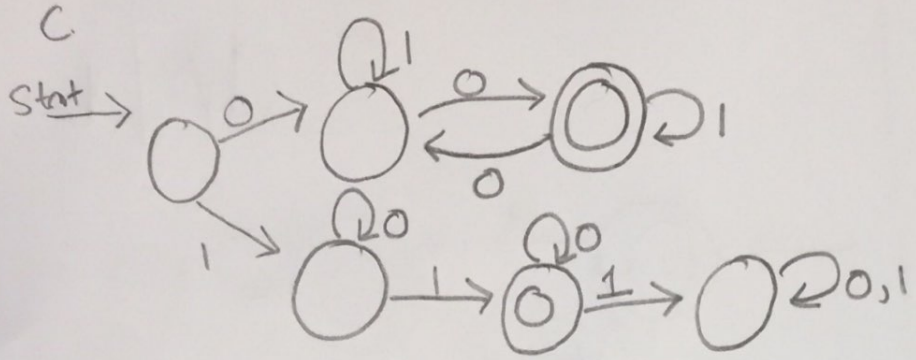
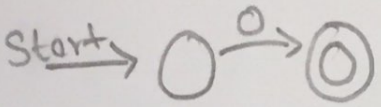
① 1.7b



c.

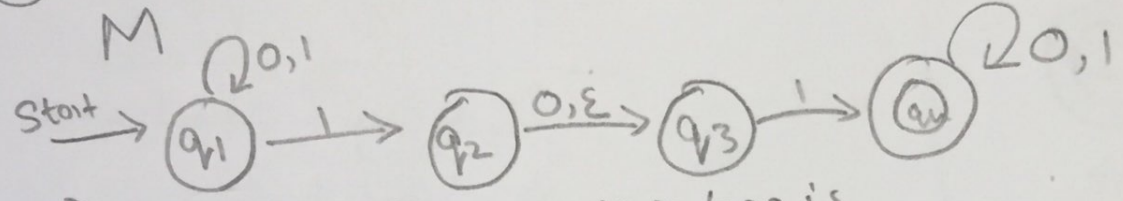


d.



② 1.14 b

M

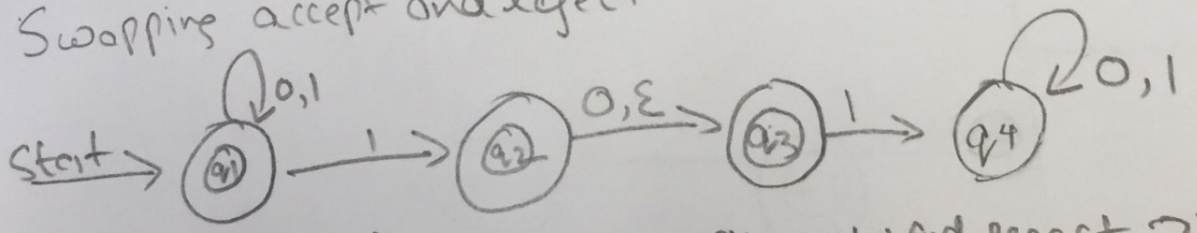


L(C) = Language that accepts 2 too is.

C = 11

M accepts this language.

Swapping accept and reject



this should accept complement; and reject 2 too is.

However, as you can see, ~~this~~ 11 is accepted.

We know that the language of a DFA is closed. ∴ if we can make a NFA, under complement the language will be closed since any NFA can be converted to a DFA.

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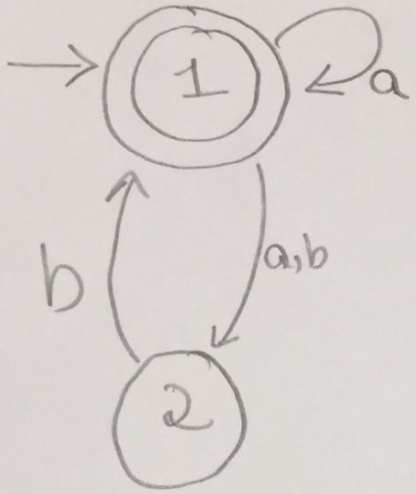
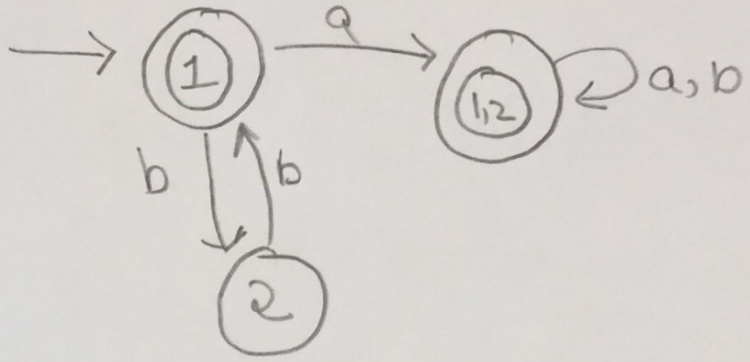
1.16

(a) Given

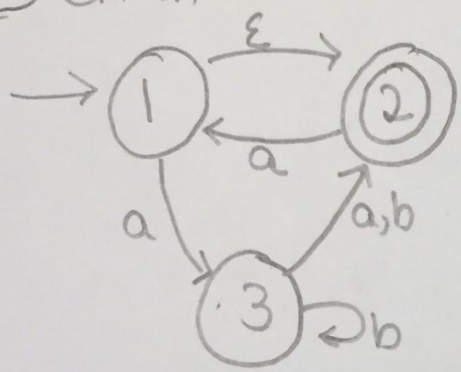


Soln (a)

	a	b
1	{1,2}	2
{1,2}	{1,2}	{1,2}
{2}	ϕ	1

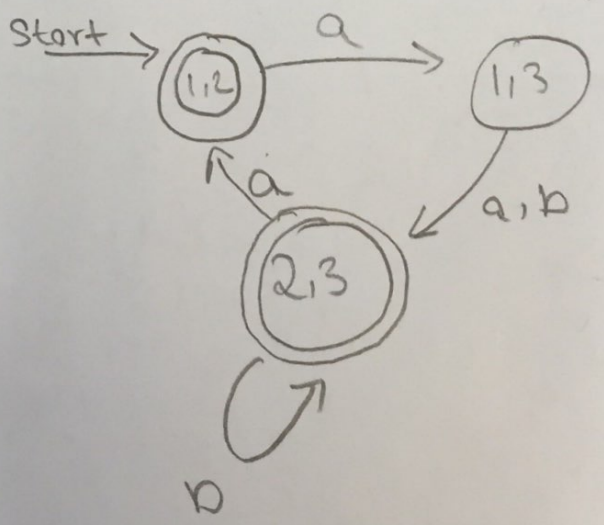


(b) Given



Soln

	a	b	ε
1	3	ϕ	{1,2}
3	2	{2,3}	3
2	1	ϕ	2
{1,2}	{1,3}	ϕ	{1,2}
{2,3}	{1,2}	{2,3}	{2,3
{1,3}	{3,2}	{2,3}	{1,3}



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(4.)

1.20 (a)

Member ① abb ② aabb

Non-member ① ba ② bb aa

(b) Member ① a ba b ② a ~~b~~ baba b

Non-member ① bbaa ② bb ba aa

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